Language Modeling

Before studying language models, we must define what one actually is:

- Formal Definition: A language model (LM) is a probabilistic mechanism for generating text, or equivalently, a probabilistic model that defines a probability distribution over sequences of text.
- Predictive View: Formulated first by Claude Shannon ("Prediction and entropy of printed English", 1951), it investigates the fundamental problem of how predictable the next word is given preceding context.

Shannon's Game

To demonstrate the probabilistic nature of language, Claude Shannon designed a letter-prediction experiment where a subject must predict the next letter of a hidden text based solely on the preceding context.

Experiment Examples:

- Target: THE ROOM WAS NOT VERY LIGHT A SMALL OBLONG
- Guesses: —-ROO——NOT-V—-LI——SM—OBL—
- Target: READING LAMP ON THE DESK SHED GLOW ON
- Guesses: REA———O—-D—-SHED-GLO–O–
- Target: POLISHED WOOD BUT LESS ON THE SHABBY RED CARPET
- Guesses: P-L-S——O—BU–L-S–O——SH—-RE–C——
- (A dash - represents a correct prediction on the first try; a revealed letter indicates an incorrect guess where the correct character had to be provided.)

Language Modeling: Questions & Implications

- Information & Unpredictability: Shannon's game provides a way to estimate the unpredictabilityand thus the information content (entropy)of each subsequent letter and word.
- Can Computers Do the Same? Testing whether computational models can match or exceed human predictive capability offers a benchmark for evaluating language technology.
- Utility: If computers can perform this prediction, what practical tasks does language modeling actually enable?
- Psychological Reality: Is there cognitive and psychological evidence that human readers and listeners are sensitive to the predictability of upcoming words? Yes, there is.

The Language Modeling Problem

Formally, building a language model means finding a probability distribution P over sequences of words. Given a prefix sequence $x_1, \ldots, x_{i-1}$, the model predicts the probability of the next word:

$$P(x_i \mid x_1, \ldots, x_{i-1})$$

More formally:

- Setup: Assume a finite set of words called the vocabulary V.
- Data: A training set of examples consisting of finite strings over V (i.e., sequences in V^*).
- Problem: Estimate a probability distribution P over all valid finite strings $x \in V^*$ such that: $\sum_{x \in V^*} P(x) = 1, \quad P(x) \geq 0$
- This formal setup allows us to apply standard probability laws to transition between joint and conditional distributions:
- Joint to Conditional (Definition of Conditional Probability):
- $P(x_n \mid x_1, \ldots, x_{n-1}) = \frac{P(x_1, \ldots, x_n)}{P(x_1, \ldots, x_{n-1})}$
- Conditional to Joint (The Chain Rule of Probability):
- $P(x) = P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid x_1, \ldots, x_{i-1})$

Applications of Language Modeling

1. Automatic Speech Recognition (ASR) & The Noisy Channel Model

In speech recognition, input acoustics a must be mapped to an output text sentence w. State of the art exhibits varying error rates depending on domain noise: $\sim 0.3\%$ for digit strings, $\sim 5\%$ for clear dictation, but 50%+ for noisy TV audio.

Because different word sequences sound identical or nearly identical (e.g., "Wreck a nice beach" vs. "Recognize speech", or "Eye eight uh Jerry" vs. "I ate a cherry"), acoustics alone cannot uniquely determine the message.

This is resolved using the Noisy Channel Model:

- Objective: Find the most likely word sequence w^* given acoustic signal a: $w^* = \arg\max_w P(w \mid a)$
- Bayes' Rule Derivation:
- $$\begin{aligned} w^* &= \arg\max_w P(w \mid a) \\ &= \arg\max_w \frac{P(a \mid w)P(w)}{P(a)} \\ &= \arg\max_w \underbrace{P(a \mid w)}_{\text{Acoustic Model}} \cdot \underbrace{P(w)}_{\text{Language Model}} \end{aligned}$$
  - Role of the Components: $P(a \mid w)$ (Acoustic Model): Scores how likely the acoustic signal matches the hypothesized words.
  - $P(w)$ (Language Model): Acts as a prior distribution over valid English sentences, penalizing nonsensical candidates even if they score highly acoustically.

2. Text Generation & Prompting

- Evolution: Language modeling originally served as a linguistic prior P(w) inside pipelines like ASR and Machine Translation. With the rise of large language models (LLMs, accelerated by GPT-2 in 2019), LMs transformed into general-purpose text generators and foundational solutions across NLP.
- Prompting: Tasks such as question answering (e.g., Q: how big is France? A: 543,940 km^2) or summarization are reframed as next-token prediction.
  - Prompt: A text prefix formatted so that the most likely continuation is the intended output.
  - Prompt Engineering: Bypasses additional model trainingthe underlying parameters remain unchanged while only the input prefix varies.

Nature of the Distribution: What Does P(x) Capture?

- Is There a "Ground Truth" Distribution? It is unclear whether an objective, true distribution over language exists; P(x) inherently varies across domains, context, genres, and individual speakers.
- Plausibility (The Classic View): At minimum, probabilities should reflect semantic and syntactic plausibility, not merely grammaticality: $P(\text{I saw a van}) \gg P(\text{eyes awe of an})$
- $P(\text{artichokes intimidate zippers}) \approx 0$
- The Scalar Reduction Trade-off:
  - Strength: Collapsing the complexity of natural language into a single scalar probability P(x) simplifies mathematical formulation and computation.
  - Weakness: Language is inherently multidimensional; projecting nuance, intent, and pragmatics onto a 1D scalar score discards rich linguistic structure.
- Density Estimation Perspective:
  - Input: A collection of observed training sentences $x \in V^*$.
  - Output: A learned system computing valid probabilities P(x).
  - Formulation: Language modeling can be framed straightforwardly as statistical density estimation over the space of finite strings drawn from a vocabulary V.

Simpler Language Models

Before training complex neural models on expensive hardware, it is useful to first look at simpler statistical baselines:

The Trivial (Empirical) Language Model

The simplest option is to predict based on the empirical distribution over the training set:

$$P(x_1, \ldots, x_n) = \frac{C(x_1, \ldots, x_n)}{N}$$

for sentence $x = x_1, \ldots, x_n$ (where C is sentence count and N is total sentences).

- Flaw: It does not generalize at all; unseen sentences receive zero probability, even though valid sentences outside the training set must retain non-zero probability.

Unigram Model

This model has a generative basis where words are drawn independently until a stop symbol is selected:

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i)$$

- Flaw: Complete disregard for context and word order.

Bigram Model

In this model, word generation is conditioned on the single preceding word, incorporating minimal local context.

- Generative Process: Pick a start symbol, sample each subsequent word contingent on the word before, and repeat until choosing stop.
- Factorization:
- $P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid x_{i-1})$

This is equivalent to assuming that the words of the sentence form a homogeneous first-order Markov chain, namely:

$$P(x_i \mid x_{i-1}, \ldots, x_0) = P(x_i \mid x_{i-1})$$

Markov Model / Markov Chain

- Definition: A finite state automaton with probabilistic state transitions.
- The Markov Assumption: The next state depends only on the current state and is conditionally independent of previous history given the current state.

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Limitations of Bigram Models: Long-Distance Dependencies

Because bigram models only consider the single immediately preceding word, they fail on long-range syntactic and semantic agreements. For example, a bigram model will likely assign similar probabilities to:

- Long-distance subject-verb agreement:
  - "The birds I saw last night sing beautifully"
  - "The birds I saw last night sings beautifully"
  - (Conditioned only on "night", both "sing" and "sings" look equally plausible).
- Anaphora and gender agreement:
  - "He saw himself in the mirror"
  - "He saw herself in the mirror"
  - (Conditioned only on "saw", "himself" and "herself" are indistinguishable without remembering "He").

Higher-Order Markov Language Models (k-gram Models)

To capture longer context, a k-th order Markov model conditions each word on the preceding k tokens:

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p(x_i \mid x_{i-1}, \ldots, x_{i-k})$$

- Equivalence to Bigram Models: A k-gram model can be viewed as a standard bigram model over an augmented state space where each state is a k-tuple of words.
- Transition Constraint: Transitions between k-tuples are constrained: to transition from state $(w_1, \ldots, w_k)$ to state $(u_1, \ldots, u_k)$, the first k-1 coordinates of the new state must match the last k-1 coordinates of the preceding state ($u_j = w_{j+1}$ for $j = 1, \ldots, k-1$).
- The Core Limitation (Parameter Explosion):
  - The number of parameters grows exponentially with k.
  - A k-gram model over a vocabulary V requires approximately $|V|^k$ entries in its transition matrix, leading to extreme data sparsity.

Formal Representation & Learning of Markov LMs

The parameters of a Markov LM form a positive-valued, row-stochastic matrix A (the transition matrix):

$$A_{ij} = P(x_m = w_j \mid x_{m-1} = w_i) \quad \text{where } w_i, w_j \in V \cup \{\text{START}, \text{STOP}\}$$

- Boundary Markers: Sequences are treated as starting with instances of a special START symbol and ending with a terminal STOP symbol.

Transition Matrix Example

For a small vocabulary {FUNNY, LITTLE, THINGS} plus boundary tokens:

$$A = \begin{pmatrix} 0 & 0.5 & 0.5 & 0 & 0 \\ 0 & 0.8 & 0.1 & 0.1 & 0 \\ 0 & 0.1 & 0.1 & 0.8 & 0 \\ 0 & 0 & 0 & 0.1 & 0.9 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

(Rows represent $w_{m-1} \in \{\text{START}, \text{FUNNY}, \text{LITTLE}, \text{THINGS}, \text{STOP}\}$; columns represent w_m in the same order. Each row sums to 1).

Maximum Likelihood Estimation (MLE)

The standard way to estimate the transition probabilities from training data is by empirical counting:

$$q_{ML}(x_m = w_j \mid x_{m-1} = w_i) = \frac{\text{count}(w_i, w_j)}{\sum_{j'} \text{count}(w_i, w_{j'})} = \frac{\text{count}(w_i, w_j)}{\text{count}(w_i)}$$

For a k-th order Markov language model, the Maximum Likelihood Estimator (MLE) is obtained by empirical count ratios:

$$q_{ML}(x_i = w \mid x_{i-k}, \ldots, x_{i-1}) = \frac{\text{count}(x_{i-k}, \ldots, x_{i-1}, w)}{\sum_{w' \in V} \text{count}(x_{i-k}, \ldots, x_{i-1}, w')} = \frac{\text{count}(x_{i-k}, \ldots, x_{i-1}, w)}{\text{count}(x_{i-k}, \ldots, x_{i-1})}$$

The Zero-Count Problem

Count-based models rely entirely on the training set, which frequently fails to cover valid linguistic usage. Consider the following distribution mismatch between training observations and valid test expressions:

Training Set Observations	**Test Set Observations**
... denied the allegations	... denied the offer
... denied the reports	... denied the loan
... denied the claims	
... denied the request	

Because the transition sequence "denied the offer" never appeared in training, its MLE probability is 0. Under the chain rule, a single zero probability zeros out the joint probability of the entire sentence:

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid x_{i-1}) = 0$$

Consequently, zero-probability estimates must be systematically avoided by applying smoothing techniquessome statistically grounded, others heuristicto reallocate probability mass to rare and unseen events.

Add-δ (Laplace) Smoothing

The simplest baseline adds a pseudocount $\delta > 0$ (typically $\delta = 1$ for Laplace or $\delta < 1$ for Lidstone) to every word count:

$$q_{\text{add-}\delta}(w) = \frac{c(w) + \delta}{N + \delta|V|}$$

(where $N = \sum_{w'} c(w')$ is total token count and $|V|$ is the vocabulary size).

For conditioned n-grams:

$$q_{\text{add-}\delta}(w_i \mid w_{i-1}) = \frac{c(w_{i-1}, w_i) + \delta}{c(w_{i-1}) + \delta|V|}$$

- Practical Failure: Add-δ fails in practice for large vocabularies. Because $|V|$ is large (e.g., 10^5), $\delta|V|$ in the denominator dominates the actual counts, moving far too much probability mass from observed frequent events to the vast number of unseen combinations.

Unreliable MLE Beyond Zero Counts

The core issue extends beyond exact zero counts:

- Low-Frequency Conditioning: Whenever the conditioning history itself has low frequency, the empirical count ratio (MLE) becomes statistically unreliable and exhibits high variance.
- N-gram Estimation Challenges: Estimating n-gram probabilities is a fundamental NLP task beyond language models (e.g., extracting collocations to discern word semantics). As n increases, sparsity compounds because the space of conditioning contexts grows exponentially ($|V|^{n-1}$), rendering raw MLE counts increasingly

Back-off and Linear Interpolation

Simplified Notation

Assuming transition probabilities do not depend on the sequence position m (homogeneous Markov models), we abbreviate:

$$P(w_j \mid w_i, w_l) \quad \text{instead of} \quad P(x_m = w_j \mid x_{m-1} = w_i, x_{m-2} = w_l)$$

The Back-off Principle

When higher-order conditioning context lacks sufficient statistics, we progressively fall back ("back-off") to lower-order models:

1. Trigram MLE:
$$q_{ML}(x_m = w_j \mid x_{m-1} = w_i, x_{m-2} = w_l) = \frac{\text{count}(w_l, w_i, w_j)}{\sum_{j'} \text{count}(w_l, w_i, w_{j'})}$$
2. Back-off to Bigram: If (w_l, w_i) was not observed frequently enough, condition only on w_i:
$$q_{ML}(x_m = w_j \mid x_{m-1} = w_i) = \frac{\text{count}(w_i, w_j)}{\sum_{j'} \text{count}(w_i, w_{j'})}$$
3. Back-off to Unigram: If w_i is also infrequent, fall back to the unigram frequency of w_j:
$$q_{ML}(x_m = w_j) = \frac{\text{count}(w_j)}{\sum_{j'} \text{count}(w_{j'})}$$

Linear Interpolation (Convex Combination)

Rather than hard switching between orders, linear interpolation blends trigram, bigram, and unigram estimates via a convex combination:

$$q(w_i \mid w_{i-2}, w_{i-1}) = \lambda_1 q_{ML}(w_i \mid w_{i-2}, w_{i-1}) + \lambda_2 q_{ML}(w_i \mid w_{i-1}) + \lambda_3 q_{ML}(w_i)$$

$$\text{where } \lambda_1 + \lambda_2 + \lambda_3 = 1, \quad \lambda_i \geq 0 \quad \forall i$$

Probability and Likelihood of a Corpus

- Corpus Probability: The probability of a corpus of N independent sentences $s_1, \ldots, s_N$ is the product of sentence probabilities:
$$P(\text{corpus}) = \prod_{i=1}^{N} P(s_i) = \prod_{i=1}^{N} \prod_{j=1}^{\text{length}(s_i)} P(w_j^{(i)} \mid w_{j-1}^{(i)}, w_{j-2}^{(i)})$$
- Count Formulation: If $c'(w_1, w_2, w_3)$ denotes the count of trigram (w_1, w_2, w_3) in the corpus, the product collapses to:
$$P(\text{corpus}) = \prod_{(w_1, w_2, w_3) \in V^3} P(w_3 \mid w_1, w_2)^{c'(w_1, w_2, w_3)}$$

Estimating Interpolation Weights (λ_i)

To prevent overfitting, the dataset is split into training and validation sets: compute the q_{ML} estimates on training data, then search for weights λ_i that maximize log-likelihood over the validation set.

This yields a concave optimization problem over validation counts $c'(w_1, w_2, w_3)$:

$$L(\lambda_1, \lambda_2, \lambda_3) = \sum_{w_1, w_2, w_3} c'(w_1, w_2, w_3) \log q(w_3 \mid w_1, w_2)$$

$$\text{subject to } \sum_{i=1}^{3} \lambda_i = 1 \quad \text{and} \quad \lambda_i \geq 0 \quad \forall i$$

Context-Dependent Interpolation Weights

To allow the model to trust the trigram more when the conditioning history is frequent, we can condition the λ weights on history frequency bins.

Define a binning function $\Pi(w_{i-2}, w_{i-1})$ over context counts:

$$\Pi(w_{i-2}, w_{i-1}) = \begin{cases} 1 & \text{if Count}(w_{i-1}, w_{i-2}) = 0 \\ 2 & \text{if } 1 \leq \text{Count}(w_{i-1}, w_{i-2}) \leq 2 \\ 3 & \text{if } 3 \leq \text{Count}(w_{i-1}, w_{i-2}) \leq 5 \\ 4 & \text{otherwise} \end{cases}$$

The interpolated probability becomes:

$$q(w_i \mid w_{i-2}, w_{i-1}) = \lambda_1^{\Pi} q_{ML}(w_i \mid w_{i-2}, w_{i-1}) + \lambda_2^{\Pi} q_{ML}(w_i \mid w_{i-1}) + \lambda_3^{\Pi} q_{ML}(w_i)$$
$$\text{where for each bin } \Pi: \quad \lambda_1^{\Pi} + \lambda_2^{\Pi} + \lambda_3^{\Pi} = 1, \quad \lambda_i^{\Pi} \geq 0$$

Discounting Methods

Maximum likelihood estimates (MLE) tend to overestimate the probability of low-count items. When an event is observed only once or twice, raw counting assigns it too much probability mass relative to unseen events.

Absolute Discounting Example

To correct for this, a simple approach is absolute discounting: subtracting a fixed constant d (e.g., d = 0.5) from all positive counts:

History x	**Continuations**	**Count**(x)	**Count**$^*(x)$ (Discounted)	$q_{ML}(w_i \mid w_{i-1})$	$\frac{\textbf{Count}^*(x)}{\textbf{Count(history)}}$
X	(total occurrences)	48			
X	dog	15	14.5	15/48	14.5/48
X	woman	11	10.5	11/48	10.5/48
X	man	10	9.5	10/48	9.5/48
X	park	5	4.5	5/48	4.5/48
X	job	2	1.5	2/48	1.5/48
X	telescope	1	0.5	1/48	0.5/48
X	manual	1	0.5	1/48	0.5/48
X	afternoon	1	0.5	1/48	0.5/48
X	country	1	0.5	1/48	0.5/48
X	street	1	0.5	1/48	0.5/48

(In the example above, shaving 0.5 off each of the 10 distinct observed continuation types leaves a pool of leftover probability mass).

Recovering the Missing Probability Mass

Subtracting d creates a "reserved" pool of probability mass that must be redistributed to unseen events (bigrams with probability zero).

For a given conditioning word w_{i-1}, the missing probability mass $\lambda(w_{i-1})$ is:

$$\lambda(w_{i-1}) = 1 - \sum_{w} \frac{c^*(w_{i-1}, w)}{c(w_{i-1})}$$

In the example above (10 distinct observed words, d = 0.5, total count = 48):

$$\lambda(w_{i-1}) = \frac{10 \cdot 0.5}{48} = \frac{5}{48}$$

This mass is distributed across unseen words so that $P(w_i)$ becomes non-zero for all valid vocabulary items.

Discounting + Unigram Interpolation (Kneser-Ney Formulation)

Rather than distributing the reserved mass $\lambda(w_{i-1})$ uniformly, it is interpolated with a lower-order distribution. This forms the basis of Kneser-Ney (KN) Smoothing:

$$P_{KN}(w_i \mid w_{i-1}) = \underbrace{\frac{\max\left(c(w_{i-1}, w_i) - d,\, 0\right)}{c(w_{i-1})}}_{\text{Discounted Bigram}} + \underbrace{\lambda(w_{i-1})}_{\text{Interpolation Weight}} \cdot \underbrace{P_{CONTINUATION}(w_i)}_{\text{Lower-Order Continuation Probability}}$$

- Discounted Bigram: Retains empirical evidence from the corpus while shaving off d mass from observed bigrams.
- Interpolation Weight $\lambda(w_{i-1})$: Normalizes the distribution by setting the weight exactly equal to the total mass subtracted.
- $P_{CONTINUATION}(w_i)$: Replaces the standard unigram frequency $P(w_i)$ with a continuation probability that scores how likely w_i is to complete an arbitrary, novel context.

Kneser-Ney Smoothing: Continuation Probabilities

Motivation: Flaw of Raw Unigram Frequencies

Standard interpolation uses the raw unigram probability $q_{ML}(w)$ as the fallback when higher-order counts fail. However, raw frequency does not measure how versatile a word is across diverse contexts:

- Consider predicting the blank in: "I can't see without my reading _____?"
- The word "Francisco" is far more frequent in general text than "glasses".
- However, "Francisco" almost exclusively follows the single prefix word "San".
- Using unigram frequency would assign a spuriously high probability to "reading Francisco", even though "Francisco" is not a flexible continuation word.

The Continuation Probability ($P_{CONTINUATION}$)

Instead of unigram frequency, Kneser-Ney interpolates with a metric capturing: "How likely is w to appear as a novel continuation in an unseen context?"

- Rather than counting token occurrences, we count the number of distinct bigram types word w completes (since every bigram type was a novel continuation the first time it appeared):

$$P_{CONTINUATION}(w) \propto |\{w_{i-1} : c(w_{i-1}, w) > 0\}|$$

- Properly Normalized: Dividing by the total number of distinct bigram types in the entire vocabulary yields a valid probability distribution:

$$P_{CONTINUATION}(w) = \frac{|\{w_{i-1} : c(w_{i-1}, w) > 0\}|}{|\{(w_{j-1}, w_j) : c(w_{j-1}, w_j) > 0\}|}$$

- Behavior: A word occurring frequently in only one fixed collocation (like "San Francisco") has a low continuation count ($|\{w_{i-1}\}| = 1$), resulting in a low $P_{CONTINUATION}$. Conversely, words that appear after many distinct words receive high continuation probabilities.

The Complete Kneser-Ney Bigram Formula

Combining absolute discounting with the continuation probability gives the interpolated Kneser-Ney estimator:

$$P_{KN}(w_i \mid w_{i-1}) = \frac{\max\left(c(w_{i-1}, w_i) - d,\, 0\right)}{c(w_{i-1})} + \lambda(w_{i-1}) P_{CONTINUATION}(w_i)$$

- Discount Parameter: d is a fixed discount constant, where $d \leq 1$.
- Normalization Weight $\lambda(w_{i-1})$: Matches the total probability mass discounted from observed continuations following w_{i-1}:

$$\lambda(w_{i-1}) = \frac{d}{c(w_{i-1})} \cdot |\{w : c(w_{i-1}, w) > 0\}|$$

- (where $|\{w : c(w_{i-1}, w) > 0\}|$ is the number of unique word types observed immediately after w_{i-1}).

Generalization in Language Models

A fundamental criticism of Markov language models (and classical statistical NLP more broadly, notably raised by Noam Chomsky) is their struggle to generalize beyond directly observed training data.

While smoothing provides a basic buffer against zero counts, it fails to capture syntactic structure:

- "Colorless green ideas sleep furiously" (syntactically grammatical, semantically nonsensical, zero training frequency)
- "Furiously sleep ideas green colorless" (completely ungrammatical, zero training frequency)

Under a standard n-gram model with count-based smoothing, both unseen sequences receive virtually identical, minuscule probabilities.

Pereiras Solution: Class-Based / Latent Category Models

To address this limitation, Fernando Pereira (2000) introduced a latent-variable approach:

- Latent Categories: Assume each word appearance is accompanied by an unobserved, latent category $c \in \text{CATS}$.
- Predictive Role: The identity of the next word is determined by this latent category rather than the raw lexical surface token alone.
- Distributional Clustering: Words cluster into the same category if they share similar continuation distributions.
  - Example: "the" and "a" share a category because both systematically precede singular nouns ("the/a cat", "the/a dog", "the/a university").

Formal Probabilistic Formulation

By marginalizing over the set of latent categories CATS:

$$\begin{aligned}\Pr(w_i \mid w_{i-1}) &= \sum_{c \in \text{CATS}} \Pr(w_i, c \mid w_{i-1}) \\ &= \sum_{c \in \text{CATS}} \Pr(c \mid w_{i-1}) \cdot \Pr(w_i \mid c, w_{i-1}) \\ &= \sum_{c \in \text{CATS}} \Pr(c \mid w_{i-1}) \cdot \Pr(w_i \mid c)\end{aligned}$$

- Conditional Independence Assumption: The final step assumes that the current word w_i and the preceding word w_{i-1} are conditionally independent given the latent category c.
- Optimization: The model is trained using the Expectation-Maximization (EM) algorithm, freely clustering words into categories such that category membership provides the optimal predictive signal for subsequent tokens.

Empirical Findings & Modern Relevance

Using only 16 latent categories trained via EM on newspaper text, Pereira demonstrated that statistical models can capture syntactic grammaticality without explicit rules:

$$\frac{\Pr(\text{Colorless green ideas sleep furiously})}{\Pr(\text{Furiously sleep ideas green colorless})} \approx 200{,}000$$

- Modern Connection: Contemporary language models operate on an extension of this principle: rather than discrete clusters, they map tokens to continuous, multi-dimensional representations (dense vector embeddings), allowing statistical generalization across grammatically and semantically similar contexts even when specific sequences are entirely novel.

Language Model Evaluation: Perplexity

The standard approach for intrinsic evaluation of a language model is assessing the probability it assigns to unseen held-out (test) data:

- Setup: Let $s_1, s_2, \ldots, s_m$ be a held-out test set containing m sentences with a total of M tokens (words).
- Cross-Entropy / Average Log-Likelihood: The average negative log probability per word across the test corpus is:

$$l = -\frac{1}{M} \sum_{i=1}^{m} \log_2 P(s_i) = -\frac{1}{M} \log_2 P(\text{held-out data})$$

- Perplexity (PPL): Formally defined as 2 raised to the cross-entropy:

$$\text{PPL} = 2^l = \left(\prod_{i=1}^{m} P(s_i)\right)^{-\frac{1}{M}} = \sqrt[M]{\frac{1}{P(s_1, s_2, \ldots, s_m)}}$$

- (Lower perplexity indicates higher probability assigned to the test text and thus a superior model).

Interpreting Perplexity Values

- Branching Factor: Perplexity intuitively represents the effective branching factorthe average number of equally likely candidates the model is choosing among for the next word.
- Uniform Baseline: If a model predicts uniformly across a vocabulary of size N ($P(w) = \frac{1}{N}$ regardless of context), its perplexity is exactly N.
- Empirical Ranges:
  - Perplexity depends heavily on domain, vocabulary size, and text genre.
  - Classical Markov (n-gram) models typically yield perplexity values between 50 and 500.
  - Modern neural and transformer architectures achieve values an order of magnitude or more lower (often sub-20 on standard benchmarks).
- Extrinsic Evaluation: Because intrinsic perplexity does not always correlate perfectly with end-task performance, models are routinely evaluated extrinsically by plugging them into downstream pipelines (e.g., ASR accuracy, machine translation BLEU, or classification tasks).

Predictability and Psycholinguistics

Language modeling is deeply connected to human cognitive language processing and psycholinguistic findings:

- Reading Time: Word-by-word reading time (measured via eye-tracking or self-paced reading) is logarithmic in the words predictability / surprisal ($-\log P(w \mid \text{context})$) (Smith and Levy, 2013).
- Word Length: Lexical length across natural languages correlates with average information content rather than frequency alone; words carrying less information in context tend to be shorter (Piantadosi et al., 2010).
- Uniform Information Density (UID): The UID hypothesis posits that speakers optimize communication by distributing information evenly across an utterance, avoiding both abrupt information spikes (cognitive overload) and excessive troughs (inefficiency).
- Electrophysiological Evidence (ERPs): Brain responses measured via Event-Related Potentials show clear electrical spikes when prediction is violatedmost notably the N400 wave, whose amplitude scales directly with how improbable or semantically unexpected an upcoming word is.